

6 Competitive Equilibrium

A competitive equilibrium is a sequence of prices and allocations: $\{c_t^*, K_t^{s*}, K_t^{d*}, n_t^{s*}, n_t^{d*}\}_{t \geq 0}$ and prices $\{R_t^*, w_t^*\}_{t \geq 0}$.

1. Utility maximization

$$\begin{aligned} \{c_t^*, K_{t+1}^{s*}, n_t^{s*}\}_{t \geq 0} &\in \arg \max \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{subject to } c_t + K_{t+1} &= R_t^* K_t + w_t^* n_t \\ K_0^{s*} &= K_0 \text{ is given} \end{aligned}$$

2. Profit maximization

$$K_t^{d*}, n_t^{d*} \in \arg \max_{K, n} F(K, n) + (1 - \delta)K - w_t^* n - R_t^* K$$

$$3. n_t^{s*} = n_t^{d*}$$

$$4. K_t^{s*} = K_t^{d*}.$$

Consider the budget set of the household:

$$\sum_0^T \frac{c_t}{\Pi_{s=1}^t R_s^*} + \frac{K_{T+1}}{\Pi_{s=1}^T R_s^*} = \sum_0^T \frac{w_t}{\Pi_{s=1}^t R_s^*} + R_0^* K_0$$

Under non-ponzi-game condition

$$\lim_{T \rightarrow \infty} \frac{K_{T+1}}{\Pi_{s=1}^T R_s^*} \geq 0,$$

the budget constraint in the limit is

$$\sum_0^{\infty} \frac{c_t}{\Pi_{s=1}^t R_s^*} \leq \sum_0^{\infty} \frac{w_t}{\Pi_{s=1}^t R_s^*} + R_0^* K_0$$

LEMMA 2 *An allocation solves the household's problem if and only if*

$$u'(c_t) = \beta u'(c_{t+1}) R_{t+1}^*, \forall t$$

$$\sum_0^\infty \frac{c_t}{\Pi_{s=1}^t R_s^*} = \sum_0^\infty \frac{w_t}{\Pi_{s=1}^t R_s^*} + R_0^* K_0.$$

The latter equation is the same as the transversality condition of the household

$$\lim_{T \rightarrow \infty} \frac{K_{T+1}}{\Pi_{s=1}^T R_s^*} = 0. \text{ (present value of the long-run savings is zero)}$$

THEOREM 8 (SECOND WELFARE THEOREM) *If an allocation is pareto efficient (i.e., solve the social planner's problem), then we can find prices such that the prices and the allocation constitute an equilibrium.*

THEOREM 9 (FIRST WELFARE THEOREM) *The allocation in a competitive equilibrium solves the social planner's problem (it is pareto efficient).*

PROOF OF : Suppose the equilibrium allocation is

$$K_0, K_1, K_2 \dots$$

household maximization implies

$$\lim_{T \rightarrow \infty} \frac{K_{T+1}}{\Pi_{s=1}^T R_s^*} = 0.$$

Because $\Pi_{s=1}^T R_s^* = \frac{u'(c_0)}{\beta^T u'(c_t)}$, we get

$$\lim_{T \rightarrow \infty} \frac{\beta^T u'(c_t)}{u'(c_0)} K_{T+1} = 0,$$

which is the sufficient condition for the social planner's problem.

Q.E.D.